

ALFAA14499

## Barium hydroxide octahydrate

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
 Product Description: 八水合氢氧化钡  
 Barium hydroxide octahydrate

Cat No. : 14499  
 CAS No 12230-71-6  
 Molecular Formula H<sub>2</sub> Ba O<sub>2</sub> · 8 H<sub>2</sub> O

Supplier Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
 Shore Road, Heysham  
 Lancashire, LA3 2XY,  
 United Kingdom  
 Office Tel: +44 (0) 1524 850506  
 Office Fax: +44 (0) 1524 850608

Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

Physical State  
 Solid

Appearance  
 White

Odor  
 Odorless

#### Emergency Overview

Causes severe skin burns and eye damage. May cause damage to organs. May cause respiratory irritation. Harmful if swallowed.  
 Harmful if inhaled. Air sensitive.

#### Classification of the substance or mixture

|                                                    |                       |
|----------------------------------------------------|-----------------------|
| Acute Oral Toxicity                                | Category 4            |
| Acute Inhalation Toxicity - Dusts and Mists        | Category 4            |
| Skin Corrosion/Irritation                          | Category 1 A          |
| Serious Eye Damage/Eye Irritation                  | Category 1            |
| Specific target organ toxicity - (single exposure) | Category 2 Category 3 |

#### Label Elements



## Barium hydroxide octahydrate

## Signal Word

## Danger

## Hazard Statements

H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation  
H371 - May cause damage to organs  
H302 + H332 - Harmful if swallowed or if inhaled

## Precautionary Statements

## Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P331 - Do NOT induce vomiting  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

None identified.

## Health Hazards

Corrosive. Causes skin and eye burns. Causes serious eye damage. May cause damage to organs. May cause respiratory irritation. Harmful if swallowed. Harmful if inhaled.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

## Other Hazards

May cause cardiac arrhythmia  
Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                    | CAS No     | Weight % |
|------------------------------|------------|----------|
| Barium hydroxide octahydrate | 12230-71-6 | >95      |
| Barium hydroxide (Ba(OH)2)   | 17194-00-2 | -        |

## Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.  
Keep eye wide open while rinsing.

## Barium hydroxide octahydrate

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Ingestion**

Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.

**Most important symptoms and effects**

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

The product causes burns of eyes, skin and mucous membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

**Environmental Precautions**

Should not be released into the environment. Do not allow material to contaminate ground water system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume

## Barium hydroxide octahydrate

hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep container tightly closed. Keep in a dry, cool and well-ventilated place. Keep under nitrogen. Corrosives area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component                               | China | Taiwan                     | Thailand                   | Hong Kong |
|-----------------------------------------|-------|----------------------------|----------------------------|-----------|
| Barium hydroxide octahydrate            | -     | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | -         |
| Barium hydroxide (Ba(OH) <sub>2</sub> ) | -     | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.5 mg/m <sup>3</sup> | -         |

| Component                               | ACGIH TLV                  | OSHA PEL                             | NIOSH                                                    | The United Kingdom                                                    | European Union                   |
|-----------------------------------------|----------------------------|--------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------|
| Barium hydroxide octahydrate            | TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr | TWA: 0.5 mg/m <sup>3</sup> (8hr) |
| Barium hydroxide (Ba(OH) <sub>2</sub> ) | TWA: 0.5 mg/m <sup>3</sup> | (Vacated) TWA: 0.5 mg/m <sup>3</sup> | IDLH: 50 mg/m <sup>3</sup><br>TWA: 0.5 mg/m <sup>3</sup> | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr | TWA: 0.5 mg/m <sup>3</sup> (8hr) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material                                    | Breakthrough time | Glove thickness | EU standard | Glove comments                                                                 |
|---------------------------------------------------|-------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Latex gloves<br>Nitrile rubber<br>Neoprene<br>PVC | > 480 minutes     | 0.6 mm          | EN 374      | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

## Barium hydroxide octahydrate

|                                        |                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                  |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particulates filter conforming to EN 143                                                          |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Particle filtering: EN149:2001<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                   |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                            |                                          |
|------------------------------------------------|----------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                      |                                          |
| <b>Physical State</b>                          | Solid                      |                                          |
| <b>Odor</b>                                    | Odorless                   |                                          |
| <b>Odor Threshold</b>                          | No data available          |                                          |
| <b>pH</b>                                      | 12.5                       |                                          |
| <b>Melting Point/Range</b>                     | 78 °C / 172.4 °F           |                                          |
| <b>Softening Point</b>                         | No data available          |                                          |
| <b>Boiling Point/Range</b>                     | No information available   |                                          |
| <b>Flash Point</b>                             | No information available   | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable             | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available   |                                          |
| <b>Explosion Limits</b>                        | No data available          |                                          |
| <b>Vapor Pressure</b>                          | No data available          |                                          |
| <b>Vapor Density</b>                           | Not applicable             | Solid                                    |
| <b>Specific Gravity / Density</b>              | 2.180                      |                                          |
| <b>Bulk Density</b>                            | ca. 1.07 kg/m <sup>3</sup> |                                          |
| <b>Water Solubility</b>                        | 947 g/l                    |                                          |
| <b>Solubility in other solvents</b>            | No information available   |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                            |                                          |
| <b>Autoignition Temperature</b>                | No data available          |                                          |
| <b>Decomposition Temperature</b>               | No data available          |                                          |
| <b>Viscosity</b>                               | Not applicable             | Solid                                    |
| <b>Explosive Properties</b>                    | No information available   |                                          |
| <b>Oxidizing Properties</b>                    | Not oxidising              |                                          |
| <b>Molecular Formula</b>                       | H2 Ba O2 . 8 H2 O          |                                          |
| <b>Molecular Weight</b>                        | 315.48                     |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                |
|---------------------------------|------------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions. Air sensitive. |
| <b>Hazardous Reactions</b>      | None under normal processing.                  |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.       |

## Barium hydroxide octahydrate

**Conditions to Avoid** Exposure to air. Incompatible products. Exposure to moist air or water.**Materials to avoid** Acids. Metals. Aluminium.**Hazardous Decomposition Products** Barium oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component                               | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|-----------------------------------------|--------------------------|-------------|-----------------|
| Barium hydroxide octahydrate            | 550 mg/kg (Rat)          |             |                 |
| Barium hydroxide (Ba(OH) <sub>2</sub> ) | LD50 = 308 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

## (d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; Not applicable  
Solid**Other Adverse Effects** May cause cardiac arrhythmia**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Large amounts will affect pH and harm aquatic organisms.

## Barium hydroxide octahydrate

**Persistence and Degradability****Persistence**

Soluble in water, Persistence is unlikely, based on information available.

**Degradability**

Not relevant for inorganic substances.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with high pH-value must be neutralized before discharge.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport****UN-No**

UN3262

**Proper Shipping Name**

Corrosive solid, basic, inorganic, n.o.s.

**Technical Shipping Name**

Barium hydroxide octahydrate

**Hazard Class**

8

**Packing Group**

II

**IMDG/IMO****UN-No**

UN3262

**Proper Shipping Name**

Corrosive solid, basic, inorganic, n.o.s.

**Technical Shipping Name**

Barium hydroxide octahydrate

**Hazard Class**

8

**Packing Group**

II

**IATA****UN-No**

UN3262

**Proper Shipping Name**

Corrosive solid, basic, inorganic, n.o.s.

**Technical Shipping Name**

Barium hydroxide octahydrate

**Hazard Class**

8

**Packing Group**

II

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION**

## Barium hydroxide octahydrate

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Barium hydroxide octahydrate            | -                                                   | X                                       | X    | X     | -         | -    | -   | X     | -    |      | X    | -        |
| Barium hydroxide (Ba(OH) <sub>2</sub> ) | X                                                   | X                                       | X    | X     | 241-234-5 | X    | X   | X     | X    | X    | X    | KE-02060 |

## Note

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

## National Regulations

## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 06-Nov-2010  
**Revision Date** 06-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate



**Barium hydroxide octahydrate**

---

**BCF** - Bioconcentration factor**VOC** - (Volatile Organic Compound)**Key literature references and sources for data**<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**